

MOBILITY ANALYSIS FOR EXCEPTIONAL EVENTS – Data Delivery Guide

The data you're going to work with are mobility data derived from mobile phone data: movements are tracked through the interaction of devices with phone cells, processed by algorithms and anonymized. This data is in fact a secondary product of telephony, which is why on one hand they can be very useful for analyzing people's mobility as they are a large amount of informations collected almost in real time, on the other hand they need to be handled and interpreted with care.

Using the datasets provided, you can work on origin-destination relationships, focusing on flows involving Campania, both within the region and in relation to the rest of Italy.

This will allow you to observe which areas could become centers of attraction, which areas could serve as places of accommodation or transit, and how the presence of visitors in accommodation facilities could translate into daily movements towards the Gulf locations where the regattas and side events will be held. The analysis you carry out will therefore serve to answer questions such as: what will happen to the flow of people during the America's Cup? Where will the demand for mobility be concentrated? Is the current system capable of absorbing it or is it at risk of collapse?

A possible work plan could be as follows:

1. Analysis of the services currently offered, with particular attention to public services and transport. The aim is to understand how the system works today, what connections exist, which areas are already well served and which ones are critical, because only by knowing the starting point is it possible to assess the impact of an extraordinary event.
2. Estimate the increase in demand linked to the America's Cup by comparing it with similar cases in the past, such as other editions of the event or major international sporting events: here you need data to quantify and localise the increase in flows, distinguishing between movements within Campania and flows from outside the region.
3. Use the results of previous analyses to consider how to expand, modulate or enhance existing services.

O-D 3000 ZONE MATRIX DATASET STRUCTURE

- **OD_3000_zone_202410_matrice_fondamentale.csv** = fundamental matrix containing O-D information.
- **OD_3000_zone_202410_matrice_integrativa.csv** = integrative matrix containing O-D information.
- **shape file** = spatial information for constructing maps. It also contains the names of the areas of origin/destination, associated with their specific codes.
- Lookup tables:
 1. **Lookup_giornimese.csv** = to find out the average daily journeys, you need to know how many days there are in that specific month. This table contains the day of the week, how many times that day occurs in the month (Days in Month) and an indicator for the type of day (holiday 1 or working day 0), as well as the reference year and month
 2. **vfa-ferservizi_classe_residenza.csv** = decoding file for the field "CLASSE_RESIDENZA_ID"
 3. **vfa-ferservizi_fasce_orarie_macro_lookup.csv** = decoding file for the field "FASCIA_ORARIA_MACRO_ID"
 4. **vfa-ferservizi_giorno_settimana_lookup.csv** = decoding file for the field "GIORNO_SETTIMANA_ID"
 5. **vfa-ferservizi_modalita_lookup.csv** = decoding file for the field "MODALITA_ID"
 6. **vfa-ferservizi_sistematicita_lookup.csv** = decoding file for the field "SISTEMATICITA_ID"
 7. **vfa-ferservizi_tipo_giorno_lookup.csv** = decoding file for the field "TIPO_GIORNO_ID"
 8. **vfa-ferservizi_tipo_nazionalita_lookup.csv** = decoding file for the field "TIPO_NAZIONALITA_ID"

The reference period is **October 2024**.

The representation by zone is a type of geographical representation based on the division of the Italian territory into 3,000 zones, each of which is approximately the same size as a municipality, except for large population centres where a single municipality may be divided into several zones. Conversely, smaller and less populated municipalities have been grouped into larger zones.

In the attached dataset, all zones not belonging to the Campania region have been aggregated under the heading "Italy", while details of the individual zones of the Campania provinces are available to identify internal movements within the region to the zones concerned. Both the matrices show all movements originating and directed within the region, excluding movements that do not involve them.

Origin Region	Destination Region	
	Campania	Italia
Campania	176.269.078	5.501.910
Italia	5.490.827	

The available fields are:

- YEAR = reference year (2024)
- MONTH = reference month (10)
- ORIGIN_ZONE = contains the codes of the areas of origin within the Campania region and the word 'Italia' for areas that do not belong to the Campania region
- DESTINATION_ZONE = contains the codes of the destination areas within the Campania region and the word 'Italy' for areas that do not belong to the Campania region
- CLASSE_RESIDENZA_ID = identification code of the field containing the residence of the user making the trip (resident in the area of origin, resident in the area of destination, resident elsewhere)
- FASCIA_ORARIA_MACRO_ID = identification code of the field containing the time slot in which the trip takes place
- GIORNO_SETTIMANA_ID = identification code of the field containing the day of the week on which the trip takes place
- SISTEMATICITA_ID = Identification code of the field containing information on how often the user travels to the specified destination. If they travel to that area more than 12 times a month, it is considered regular
- TIPO_GIORNO_ID = identification code of the field containing the indication of the type of day of the week (working day/public holiday)
- TIPO_NAZIONALITA_ID = identification code of the field containing the user's nationality
- MODALITA_ID = Identification code of the field containing the means of transport used for travel, as determined by standardised algorithms. Only trains and aeroplanes are recorded; all other means of transport are classified as "other".
- TRIPS = number of monthly journeys with the specific characteristics described in the other columns.

To protect user privacy, when the number of journeys (<15 units) in a certain class is very low, some columns in that class may be hidden, otherwise the user could potentially be identified by the journey made and its characteristics.

Not all measures are obscured in the same way and at the same time; there is a hierarchy among them, so that some, considered less relevant, are more subject to masking than others.

The fields are progressively masked until the minimum threshold of 15 trips is reached.

To limit the effect of masking, the dataset has been divided into two sub-matrices, which have different fields. The fields and the hierarchy between them are shown below.

Matrice fondamentale

- ANNO
- MESE
- COD_ORIGINE
- COD_DESTINAZIONE
- TIPO_GIORNO_ID
- **MACRO FASCIA ORARIA**
- CLASSE_RESIDENZA_ID
- GIORNO_SETTIMANA_ID
- TIPO_NAZIONALITA_ID
- SISTEMATICITA_ID
- VIAGGI MENSILI

Matrice integrativa

- ANNO
- MESE
- COD_ORIGINE
- COD_DESTINAZIONE
- **MODALITA_ID**
- TIPO_GIORNO_ID
- **MACRO FASCIA ORARIA**
- GIORNO_SETTIMANA_ID
- CLASSE_RESIDENZA_ID
- VIAGGI MENSILI

Soglia di stazionarietà: 1h

Mascheramento: l'ordine dei campi riflette l'ordine di «importanza» (dal meno al più mascherato)

Macro fascia oraria: 00:00-04:59/05:00-08:59/09:00-12:59/13:00-15:59/16:00-19:59/20:00-23:59

The attached spatial file contains the geometries of the 3,000 zones.

The two tables are joined using the ORIGIN_ZONE and DESTINATION_ZONE fields and the AREA_ID field of the spatial file. Due to the structure of the dataset, only the codes relating to the Campania region will match. The names of the zones can be found in the ZONA_NAME field.

The join with the *Lookup_giornimese* table is performed using (1) the field relating to the day of the week ID, (2) the year_month field, and (3) the day type, and allows you to determine how many occurrences of Monday there are in the month, how many Tuesdays, and so on for all days of the week.

O-D 3000 zone mat...	Operatore	esportazione_giorni...
# Tipo Giorno Id ▼	= ▼	# Holiday Id ▼
int(str([ANNO])+str ▼	= ▼	# Year Month ▼
# Giorno Settiman ▼	= ▼	# Day Of Week ▼

⊕ Aggiungi più campi

If the year and month are in two separate fields, as in the O-D matrix, to link them to Year_Month, a concatenation operation of the type `int(str([Year])+str([Month]))` is required.

For the month of October 2024, you will obtain:

Giorno Settimana	Giorno Settimana Id	Days In Month
Domenica	7	4
Giovedi	4	5
Lunedì	1	4
Martedì	2	5
Mercoledì	3	5
Sabato	6	4
Venerdì	5	4